

Sharp Final-Support Field–Energy Bounds for an Exact Thermal Dephasing Channel

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Abstract

The exact Weyl-displacement channel of a prescribed gapless detector is known. We solve a finite-support optimization left open after that channel is fixed. For Dirichlet half-line momentum data supported in $[0, L]$, with thermal inverse temperature β , dephasing exponent Γ , and field energy $E = \|p\|^2/2$, we prove

$$\Gamma \leq EC_\beta(L), \quad C_\beta(L) = 2L\Lambda(\pi L/\beta).$$

Here $\Lambda(\tau)$ is the simple top eigenvalue of the positive compact kernel $\pi^{-1} \log[\sinh(\tau(u+v))/\sinh(\tau|u-v|)]$ on $(0, 1)$. We give global two-sided bounds and uniform small- and large-support remainders. For the conformally coupled massless scalar in a four-dimensional de Sitter static patch, $\beta = 2\pi R$, and the same s-wave momentum profile is the unique optimizer over all angular and canonical sectors. Thus final Cauchy support $L = yR$ implies

$$\log \frac{1}{2\epsilon_{\text{obs}}} \leq E_K RC_{\text{opt}}(y), \quad C_{\text{opt}}(y) = 2y\Lambda(y/2),$$

where E_K is only the post-switch scalar Killing energy. A source radius and duration provide a causal upper envelope for L ; sharp controllability from every smaller fixed source cylinder is not claimed. Smooth compact sources approach the final-support optimum. We do not include the work or stress of an autonomous actuator or claim a total measurement-energy cost.

1 Introduction

Gapless two-level detectors linearly coupled to free fields provide an exact, nonperturbative dephasing model. Their time-ordered evolution is a Weyl displacement because free-field commutators are c-numbers. Landulfo used this structure for relativistic communication channels [1]; Barcellos and Landulfo derived the associated field-energy and switching-work accounting [2]. The same solvable detector now appears in nonperturbative treatments of Danielson–Satishchandran–Wald (DSW) decoherence [3]. Killing-horizon decoherence, its local two-point-function description, and its de Sitter specialization are also well developed [4, 5, 6, 7].

Those results determine the dephasing functional for a chosen source. We ask a different variational question: after the interaction has ended, how large can the exact thermal dephasing exponent be at fixed field energy and fixed final Cauchy support? Compact support is essential because the thermal state has arbitrarily soft modes. Localization replaces the missing spectral gap by a compact compression of the thermal inverse Hamiltonian.

We first solve this problem without imposing a relation between temperature and geometry. Let $h_0 = \sqrt{-\partial_x^2}$ on the Dirichlet half-line, let momentum data p be supported in $[0, L]$, and set

$$\Gamma_{\beta,p} = \langle p, h_0^{-1} \coth(\beta h_0/2) p \rangle, \quad E_p = \frac{1}{2} \|p\|^2. \quad (1)$$

Our first result is the exact optimum

$$\Gamma_{\beta,p} \leq E_p C_\beta(L), \quad C_\beta(L) = 2L\Lambda(\pi L/\beta), \quad (2)$$

where $\Lambda(\tau)$ is the top eigenvalue of the explicit positive compact kernel

$$k_\tau(u, v) = \frac{1}{\pi} \log \frac{\sinh[\tau(u+v)]}{\sinh[\tau|u-v|]} \quad \text{on } L^2(0, 1). \quad (3)$$

The top eigenvalue is simple and its eigenfunction is strictly positive. The answer has the global bracket

$$\max \left\{ \frac{3L}{\pi}, \frac{16L^2}{\beta\pi^2} \right\} \leq C_\beta(L) \leq \frac{4 \operatorname{arsinh}(1)}{\pi} L + \frac{16L^2}{\beta\pi^2}, \quad (4)$$

with uniform remainders that identify the vacuum-controlled small-support law and the thermal large-support law.

We then specialize to the conformally coupled massless scalar in a four-dimensional de Sitter static patch. The Bunch–Davies inverse temperature is $\beta = 2\pi R$. Optical partial waves have

$$A_\ell = -\partial_x^2 + \frac{\ell(\ell+1)}{R^2 \sinh^2(x/R)}. \quad (5)$$

Resolvent order shows that no higher angular momentum has a larger thermal momentum form. A separate localized estimate controls the field-coordinate sector, and the two lower bounds in equation (4) dominate that estimate for every $L/R > 0$. Consequently the half-line momentum optimizer is the unique optimum over the full angular and canonical phase space. For $y = L/R$,

$$C_{\text{opt}}(y) = 2y\Lambda(y/2), \quad \log \frac{1}{2\epsilon_{\text{obs}}} \leq E_K R C_{\text{opt}}(y), \quad (6)$$

where ϵ_{obs} is the halved diamond distance from complete pointer dephasing and E_K is the post-switch scalar Killing energy generated by the prescribed source. Temperature first changes the dimensionless small-support cost at cubic order, and $8y^2/\pi^3$ is the sharp leading large-support coefficient.

Sharpness in equation (6) is a statement about final Cauchy data supported in $[0, L]$. If a source is confined to areal radius $a < R$ for a static-time interval of length T , finite propagation gives only

$$L = R \operatorname{arctanh}(a/R) + T \quad (7)$$

as a causal envelope. The resulting source-radius and duration inequality is valid, but we do not claim that every fixed smaller source cylinder approaches the top eigenfunction. Smooth compact sources do approach the optimum when their worldtube approaches the final-support ball.

We do not claim a new detector channel or a new positive-kernel variational principle. The candidate contribution is the exact finite-support thermal covariance operator, its reduction from the full de Sitter angular and canonical phase space, and its sharp support asymptotics. Largest-eigenvalue and isoperimetric questions for Riesz-potential operators have an established literature [8]; whether the particular conjunction above is already implicit in that literature is the decisive novelty question.

The term field-energy requirement is literal. The quantity E_K does not include the work, clock, trap, battery, or material stress needed to implement the prescribed source. The shared smeared qubit is an idealized detector, not an autonomous relativistic probe field. Complete pointer dephasing is also only an operational channel target; the error in equation (6) is not a gravitational encoding error and is not identified with observer entropy.

Two comparisons are kept secondary. First, the dephasing form is incomparable with the local coherent-state entropy entering recent Klein–Gordon Bekenstein bounds [9]. Second, an appendix records exact flux-free Einstein–scalar final-slice constraint data for the same spherical scalar datum. That appendix does not derive the channel on a perturbed geometry or solve a coupled evolution.

Section 2 fixes the de Sitter model and recalls the exact channel. Section 3 proves the general thermal theorem and the de Sitter all-sector specialization. Section 4 proves smooth-source closure. Section 5 states the prior-art and local-entropy boundary, and section 6 summarizes the claim. Appendix A contains the optional final-slice gravity application.

2 Static-patch model and exact pointer channel

2.1 Optical partial waves

In static coordinates, four-dimensional de Sitter space has

$$ds^2 = -N(r) dt^2 + \frac{dr^2}{N(r)} + r^2 d\Omega_2^2, \quad N(r) = 1 - \frac{r^2}{R^2}, \quad 0 \leq r < R. \quad (8)$$

Introduce the optical radius

$$x = R \operatorname{arctanh}(r/R), \quad r = R \tanh(x/R), \quad \Omega(x) = \operatorname{sech}(x/R). \quad (9)$$

Then $g_{\text{dS}} = \Omega^2 g_{\text{opt}}$, where

$$g_{\text{opt}} = -dt^2 + dx^2 + R^2 \sinh^2(x/R) d\Omega_2^2. \quad (10)$$

For the conformally coupled massless scalar, spherical-harmonic decomposition and the standard radial rescaling give the positive operators

$$A_\ell = -\frac{d^2}{dx^2} + \frac{\ell(\ell+1)}{R^2 \sinh^2(x/R)}, \quad h_\ell = A_\ell^{1/2} \quad (11)$$

on $L^2(\mathbb{R}_+)$, with the regular Dirichlet condition at $x = 0$. The Bunch–Davies state restricted to the static patch is KMS at

$$\beta = 2\pi R \quad (12)$$

for the static Killing flow. The half-line theorem in section 3 first keeps β arbitrary; theorem 3.5 then imposes this de Sitter relation when comparing all angular and canonical sectors.

Let $(q_{\ell m}, p_{\ell m})$ denote real final Cauchy data for the rescaled partial waves. For smooth compact data, the thermal quadratic form and the Killing energy are

$$\Gamma_{\ell m} = \langle q_{\ell m}, h_\ell \coth(\beta h_\ell/2) q_{\ell m} \rangle + \langle p_{\ell m}, h_\ell^{-1} \coth(\beta h_\ell/2) p_{\ell m} \rangle, \quad (13)$$

$$E_{\ell m} = \frac{1}{2} \left(\|h_\ell q_{\ell m}\|^2 + \|p_{\ell m}\|^2 \right). \quad (14)$$

We write $\Gamma = \sum_{\ell m} \Gamma_{\ell m}$ and $E_K = \sum_{\ell m} E_{\ell m}$. Smooth angular data make these sums finite in the needed form domains.

For $\ell = 0$, the unitary sine transform diagonalizes h_0 . With hats denoting that transform,

$$\Gamma_{00} = \int_0^\infty \coth(\beta k/2) \left(k |\widehat{q}(k)|^2 + \frac{|\widehat{p}(k)|^2}{k} \right) dk, \quad (15)$$

$$E_{00} = \frac{1}{2} \int_0^\infty \left(k^2 |\widehat{q}(k)|^2 + |\widehat{p}(k)|^2 \right) dk. \quad (16)$$

2.2 Local interaction and exact channel

Let P be a degenerate pointer qubit with pointer observable Z_P . We couple it to the physical scalar by

$$S_{\text{int}} = - \int_{\mathcal{M}} \sqrt{-g} J(X) \phi(X) Z_P d^4 X, \quad J \in C_c^\infty(\mathcal{M}; \mathbb{R}). \quad (17)$$

The switching function and spatial smearing are both included in J . Since the commutator of two smeared free fields is a c-number, every Magnus term beyond second order vanishes. Up to a source-dependent scalar phase,

$$U_J = e^{i\Xi[J]} e^{-i\phi(J)Z_P}. \quad (18)$$

This standard exact evolution is reviewed in [1, 2, 3]. The interaction is compactly supported, and its densities commute at spacelike separation because Z_P is fixed and the scalar field is microcausal. Nevertheless, the same finite-dimensional operator Z_P appears across the smearing. Generic spatially extended nonrelativistic detectors can violate causal factorization [10]; fully relativistic localized probe fields provide a systematic replacement framework [11]. We therefore treat equation (17) as the standard prescribed gapless-detector idealization, not as an autonomous pointer field or material actuator.

Let E denote the causal propagator and K the positive-frequency map for the static one-particle structure. Tracing the Bunch–Davies field state gives a pointer dephasing channel

$$\mathcal{D}_\kappa \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a & \kappa b \\ \bar{\kappa} c & d \end{pmatrix}, \quad |\kappa| = e^{-\Gamma}, \quad \Gamma = 2 \langle KEJ, \coth(\beta h/2) KEJ \rangle. \quad (19)$$

The phase of κ is a pointer rotation and is irrelevant to the distance from complete dephasing. If $\mathcal{D}_0$ denotes complete dephasing, then

$$\epsilon_{\text{obs}} := \frac{1}{2} \|\mathcal{D}_\kappa - \mathcal{D}_0\|_\diamond = \frac{1}{2} |\kappa| = \frac{1}{2} e^{-\Gamma}. \quad (20)$$

The equality follows from $\mathcal{D}_\kappa - \mathcal{D}_0 = (\kappa/2)(\text{id} - \text{Ad}_{Z_P})$ after the correctable phase is removed, and it is saturated by a pointer superposition. The target $\mathcal{D}_0$ is the binary instance of the quantum-to-classical pointer channel used in the observer rule of Harlow, Usatyuk, and Zhao [12]. The error ϵ_{obs} measures this channel approximation only; it is not their closed-universe encoding error and is not identified with $e^{-S_{\text{ob}}}$.

After the source switches off, the two conditional field states are opposite coherent displacements. Their common excess Killing energy and excess renormalized stress are

$$E_K = \langle KEJ, hKEJ \rangle, \quad \Delta \langle T_{ab}^{\text{ren}} \rangle = T_{ab}[\phi_J]. \quad (21)$$

The state-independent Hadamard subtraction cancels in the difference, and the background one-point function vanishes. Equation (21) prices the post-switch scalar field generated by *the same* J as the channel. It does not include the stress of a clock, trap, or material device that realizes the prescribed coupling.

2.3 Causal support

Suppose J is supported inside the centered areal ball $r \leq a < R$ during a static-time interval of length T . Its optical radius is

$$\ell = R \operatorname{arctanh}(a/R). \quad (22)$$

Finite propagation in the optical metric implies that the final Cauchy data of every partial wave are supported in

$$0 \leq x \leq L, \quad L = \ell + T. \quad (23)$$

No spectral cutoff or hard spatial wall is introduced.

3 Sharp final-support thermal bounds

Let P_L be multiplication by $(0, L)$ on $L^2(\mathbb{R}_+)$. We first solve the momentum problem at arbitrary inverse temperature β . We then impose the de Sitter relation $\beta = 2\pi R$ and prove that the same momentum sector is the optimum over every angular and canonical sector.

Lemma 3.1 (Compressed half-line inverses). *For $h_0 = \sqrt{-\partial_x^2}$ with the half-line Dirichlet condition,*

$$(P_L h_0^{-1} P_L)(x, y) = \frac{1}{\pi} \log \frac{x+y}{|x-y|}, \quad (24)$$

$$(P_L h_0^{-2} P_L)(x, y) = \min(x, y). \quad (25)$$

Moreover,

$$\|P_L h_0^{-1} P_L\| \leq \frac{2 \operatorname{arsinh}(1)}{\pi} L, \quad (26)$$

$$\|P_L h_0^{-2} P_L\| = \frac{4}{\pi^2} L^2. \quad (27)$$

Proof. The unitary sine transform gives

$$\frac{2}{\pi} \int_0^\infty \frac{\sin(kx) \sin(ky)}{k} dk = \frac{1}{\pi} \log \frac{x+y}{|x-y|}$$

and

$$\frac{2}{\pi} \int_0^\infty \frac{\sin(kx) \sin(ky)}{k^2} dk = \min(x, y).$$

The logarithmic kernel is positive. With $u = x/L$, its row integral divided by L is

$$\frac{1}{\pi} [(1+u) \log(1+u) - 2u \log u - (1-u) \log(1-u)]. \quad (28)$$

Its maximum is $2 \operatorname{arsinh}(1)/\pi$ at $u = 1/\sqrt{2}$, so Schur's test proves equation (26). The kernel $\min(x, y)$ is the inverse of $-\partial_x^2$ on $(0, L)$ with Dirichlet condition at zero and Neumann condition at L . Its first eigenvalue is $(\pi/2L)^2$, proving equation (27). $\square$

Lemma 3.2 (Exact localized KMS kernel). *Set*

$$B_{\beta, L} := P_L h_0^{-1} \coth(\beta h_0/2) P_L. \quad (29)$$

Its integral kernel is

$$B_{\beta, L}(x, y) = \frac{1}{\pi} \log \frac{\sinh[\pi(x+y)/\beta]}{\sinh[\pi|x-y|/\beta]}. \quad (30)$$

Let $\tau = \pi L/\beta$ and define on $L^2(0, 1)$

$$k_\tau(u, v) := \frac{1}{\pi} \log \frac{\sinh[\tau(u+v)]}{\sinh[\tau|u-v|]}, \quad \Lambda(\tau) := \|\mathcal{K}_\tau\|. \quad (31)$$

Then $B_{\beta, L}$ is unitarily equivalent to $L\mathcal{K}_\tau$. $\mathcal{K}_\tau$ is positive, compact, and positivity improving. Its top eigenvalue $\Lambda(\tau)$ is simple and has a strictly positive eigenfunction.

Proof. Using $\coth z = 1 + 2 \sum_{n \geq 1} e^{-2nz}$ and integrating each sine-transform term gives

$$\begin{aligned} \frac{2}{\pi} \int_0^\infty \frac{\sin(kx) \sin(ky)}{k} \coth(\beta k/2) dk &= \frac{1}{\pi} \log \frac{x+y}{|x-y|} \\ &+ \frac{1}{\pi} \sum_{n \geq 1} \log \frac{(n\beta)^2 + (x+y)^2}{(n\beta)^2 + (x-y)^2}. \end{aligned} \quad (32)$$

Euler's product for $\sinh$ reduces this expression to equation (30). Scaling $x = Lu$, $y = Lv$ gives the unitary equivalence. The logarithmic diagonal singularity is square integrable on the unit square, so the operator is Hilbert–Schmidt. The spectral definition makes it positive, while the displayed kernel is strictly positive away from the measure-zero boundary and diagonal. The compact positivity-improving operator therefore has a simple positive top eigenfunction by the Perron–Frobenius–Jentzsch theorem. $\square$

Theorem 3.3 (Sharp thermal half-line momentum bound). *Let $\beta, L > 0$, let $h_0 = \sqrt{-\partial_x^2}$ on the Dirichlet half-line, and define*

$$C_\beta(L) := \sup_{\substack{p \neq 0 \\ \text{supp } p \subset [0, L]}} \frac{\langle p, h_0^{-1} \coth(\beta h_0/2) p \rangle}{\|p\|^2 / 2}. \quad (33)$$

With $\tau = \pi L/\beta$,

$$\boxed{C_\beta(L) = 2L\Lambda(\tau)}. \quad (34)$$

The normalized optimizer in the finite-energy closure is unique up to sign and is the positive top eigenfunction of $\mathcal{K}_\tau$. The global bounds

$$\max \left\{ \frac{3L}{\pi}, \frac{16L^2}{\beta\pi^2} \right\} \leq C_\beta(L) \leq \frac{4 \operatorname{arsinh}(1)}{\pi} L + \frac{16L^2}{\beta\pi^2} \quad (35)$$

hold for every $\beta, L > 0$. If

$$M(\tau) := \max_{0 < u < 1} \int_0^1 k_\tau(u, v) dv, \quad (36)$$

then $C_\beta(L) \leq 2LM(\tau)$, and the maximizing row is at

$$u_\tau = \frac{1}{\tau} \operatorname{arsinh} \left(\frac{\sinh \tau}{\sqrt{2}} \right). \quad (37)$$

Finally,

$$0 \leq C_\beta(L) - 2L\Lambda(0) \leq \frac{2\pi L^3}{3\beta^2}, \quad (38)$$

$$0 \leq C_\beta(L) - \frac{16L^2}{\beta\pi^2} \leq \frac{\beta}{2}. \quad (39)$$

Thus the exact answer depends on temperature and support only through $\tau = \pi L/\beta$, apart from its overall length scale.

Proof. By lemma 3.2, the quotient in equation (33) has supremum $2\|B_{\beta, L}\| = 2L\Lambda(\tau)$, with the stated unique optimizer. The profile $p(x) = \sqrt{3/L^3} x$ gives the vacuum lower bound $3L/\pi$, while the first eigenfunction of the Dirichlet–Neumann Green operator and $\coth z \geq z^{-1}$ give $16L^2/(\beta\pi^2)$. Conversely,

$\coth z \leq 1 + z^{-1}$ and lemma 3.1 give the upper bound in equation (35). Positivity and Schur's test give the $M(\tau)$ bound. Differentiating the row integral and using

$$\sinh[\tau(1+u)] \sinh[\tau(1-u)] = \sinh^2 \tau - \sinh^2(\tau u) \quad (40)$$

gives its unique stationary point equation (37).

For small τ , write $k_\tau = k_0 + d_\tau$. Euler's product implies, for $A \geq B \geq 0$,

$$0 \leq \log \frac{\sinh A/A}{\sinh B/B} \leq \frac{A^2 - B^2}{6}. \quad (41)$$

Taking $A = \tau(u+v)$ and $B = \tau|u-v|$ and applying Schur's test gives $0 \leq \Lambda(\tau) - \Lambda(0) \leq \tau^2/(3\pi)$, which proves equation (38). For large τ ,

$$k_\tau(u, v) = \frac{2\tau}{\pi} \min(u, v) + r_\tau(u, v). \quad (42)$$

The resolvent expansion makes r_τ nonnegative in operator order, and its explicit row integral gives $\|r_\tau\| \leq \pi/(4\tau)$. Since the norm of the $\min(u, v)$ operator is $4/\pi^2$, multiplication by $2L$ proves equation (39). $\square$

The angular potential improves the complete thermal momentum form, not only the two estimates in lemma 3.1.

Lemma 3.4 (All-angular and coordinate domination). *For every $\ell \geq 0$, $A_\ell \geq A_0$. Moreover,*

$$\begin{aligned} h_\ell^{-1} \coth(\beta h_\ell/2) &= \frac{2}{\beta} A_\ell^{-1} + \frac{4}{\beta} \sum_{n \geq 1} [A_\ell + (2\pi n/\beta)^2]^{-1} \\ &\leq h_0^{-1} \coth(\beta h_0/2) \end{aligned} \quad (43)$$

as generalized quadratic forms. If q is supported in $[0, L]$, then

$$\|q\| \leq \frac{L}{\pi} \|h_\ell q\|, \quad (44)$$

$$\langle q, h_\ell q \rangle \leq \frac{L}{\pi} \|h_\ell q\|^2, \quad (45)$$

$$\frac{\Gamma_{q, \ell m}}{E_{q, \ell m} R} \leq \frac{2y}{\pi} + \frac{2y^2}{\pi^3}, \quad y = L/R, \quad \beta = 2\pi R. \quad (46)$$

Proof. The nonnegative angular potential gives $A_\ell \geq A_0$. The Mittag-Leffler expansion $\coth z/z = z^{-2} + 2 \sum_{n \geq 1} (z^2 + \pi^2 n^2)^{-1}$ gives equation (43); every resolvent is order decreasing. Extending q by zero places it in $H_0^1(0, L)$. Dirichlet Poincare and $\|h_\ell q\|^2 = \langle q, A_\ell q \rangle$ give equation (44); Cauchy-Schwarz gives equation (45). Finally, $\coth z \leq 1 + z^{-1}$ yields

$$\Gamma_{q, \ell m} \leq \left[\frac{L}{\pi} + \frac{2L^2}{\beta\pi^2} \right] \|h_\ell q\|^2. \quad (47)$$

Divide by $E_{q, \ell m} = \|h_\ell q\|^2/2$ and set $\beta = 2\pi R$. No locality of the nonlocal vector $h_\ell q$ is assumed. $\square$

We can now specialize the half-line theorem to the static patch and prove that no angular or field-coordinate sector gives a larger quotient. The supremum below is unchanged if smooth compact data are replaced by their finite-energy closure.

Theorem 3.5 (Conformal de Sitter all-sector optimum). *Let*

$$C_{\text{opt}}(y) := \sup \left\{ \frac{\Gamma}{E_K R} : 0 < E_K < \infty, \text{supp}(q, p) \subset [0, yR] \right\}. \quad (48)$$

For the full conformally coupled massless scalar in the Bunch–Davies state, including arbitrary angular data,

$$\boxed{C_{\text{opt}}(y) = 2y\Lambda(y/2)}. \quad (49)$$

Equivalently,

$$RC_{\text{opt}}(L/R) = C_{2\pi R}(L). \quad (50)$$

The supremum lies in the s-wave momentum sector. Its normalized optimizer is the unique positive top eigenfunction of equation (31), up to an overall sign. Specializing equation (35) gives

$$\max \left\{ \frac{3y}{\pi}, \frac{8y^2}{\pi^3} \right\} \leq C_{\text{opt}}(y) \leq F(y), \quad (51)$$

$$F(y) = \frac{4 \operatorname{arsinh}(1)}{\pi} y + \frac{8}{\pi^3} y^2. \quad (52)$$

The sharper bound $C_{\text{opt}}(y) \leq 2yM(y/2)$ also follows from theorem 3.3. Its global remainders become

$$0 \leq C_{\text{opt}}(y) - 2y\Lambda(0) \leq \frac{y^3}{6\pi}, \quad (53)$$

$$0 \leq C_{\text{opt}}(y) - \frac{8y^2}{\pi^3} \leq \pi. \quad (54)$$

Thus temperature first corrects the localized vacuum cost at cubic order, while $8y^2/\pi^3$ is the sharp leading large-support coefficient.

Proof. For momentum data in the s-wave, theorem 3.3 gives

$$\sup_{p \neq 0} \frac{\Gamma_p}{E_p R} = \frac{C_{2\pi R}(L)}{R} = 2y\Lambda(y/2). \quad (55)$$

Equation (43) shows that no higher angular momentum sector is worse. The linear trial profile in equation (60) gives the lower bound $3y/\pi$. Also $\coth z \geq z^{-1}$ and equation (27) give

$$\sup_{p \neq 0} \frac{\Gamma_p}{E_p R} \geq \frac{8y^2}{\pi^3}. \quad (56)$$

The right side of equation (46) is bounded by $3y/\pi$ when $y \leq \pi^2/2$, and by $8y^2/\pi^3$ when $y \geq \pi^2/3$. These ranges overlap, so the entire coordinate sector is dominated by the s-wave momentum supremum for every $y > 0$. The domination is strict for nonzero coordinate data. Indeed, at finite β , the thermal momentum form is strictly larger than its vacuum part and strictly larger than $(2/\beta)A_0^{-1}$ on every nonzero vector. Applying those two strict inequalities to the trial profiles used above covers the same two overlapping ranges.

The angular domination is also strict for $\ell > 0$. For any $c > 0$, the resolvent difference $(A_0 + c)^{-1} - (A_\ell + c)^{-1}$ has a strictly positive quadratic form on nonzero data: equality would force $V_\ell^{1/2}(A_0 + c)^{-1}p = 0$, where $V_\ell = \ell(\ell + 1)/[R^2 \sinh^2(x/R)] > 0$ almost everywhere, and hence $p = 0$. The $n = 1$ term of equation (43) therefore excludes a higher-angular-momentum tie. Together with the simple top eigenvalue of $\mathcal{K}_{y/2}$, this proves the asserted uniqueness as well as equation (49). The explicit bracket and both remainder bounds are equations (35) and (38) and equation (39) with $\beta = 2\pi R$ and $L = yR$. $\square$

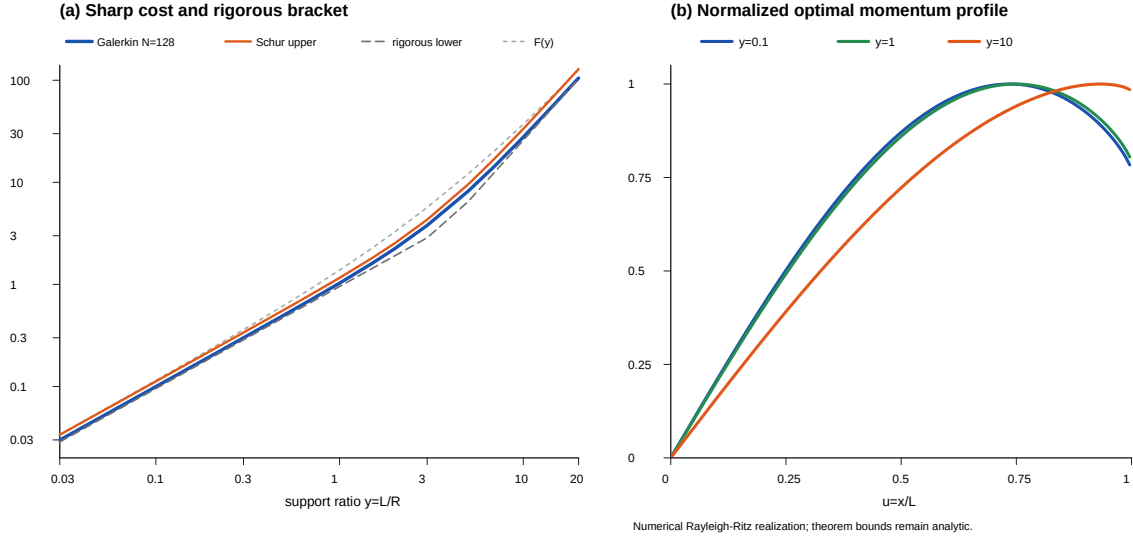


Figure 1: Numerical realization of the exact compact-kernel theorem. Panel (a) compares the 128-cell Galerkin estimate with the rigorous analytic lower bound, the exact-kernel row-Schur upper bound, and the simpler closed-form upper bound $F(y)$. Panel (b) shows the positive top eigenfunction, normalized by its maximum, at three support ratios.

3.1 Numerical realization of the sharp coefficient

The compact-operator formula can be evaluated directly without entering the proof. We use normalized piecewise constants on a uniform partition of $(0, 1)$. Cell integrals of the vacuum kernel are evaluated exactly from

$$\frac{d^2}{dz^2} \left(\frac{z^2}{2} \log |z| - \frac{3z^2}{4} \right) = \log |z|, \quad (57)$$

while Gauss–Legendre quadrature is applied only to the smooth thermal correction $\log[\sinh z/z]$. Figure 1 shows the resulting 128-cell Rayleigh–Ritz curve and positive top eigenfunctions.

At $y = 0.1, 1, 10$ the computed costs are respectively 0.0997264, 1.02960, and 27.7170. Across those three points, the largest relative change from 64 to 128 cells is 5.0×10^{-5} ; changing the thermal quadrature order from 8 to 12 at 64 cells changes the values by at most 1.1×10^{-15} . Every one of the nineteen plotted estimates lies inside equation (51) and below the exact-kernel Schur bound. Because the thermal quadrature is not interval enclosed, the plotted curve is numerical evidence rather than an additional rigorous lower bound; none of the theorem statements depends on it.

Combining equations (20), (23) and (49) gives the operational form.

Corollary 3.6 (Causal source-cylinder envelope). *If the source in equation (17) is supported in $r \leq a < R$ for a static-time interval of length T , set*

$$y = \operatorname{arctanh}(a/R) + T/R. \quad (58)$$

Then

$$\log \frac{1}{2\epsilon_{\text{obs}}} \leq E_K R C_{\text{opt}}(y) = 2E_K R y \Lambda(y/2) \leq E_K R F(y). \quad (59)$$

The first coefficient is sharp for the relaxed problem in which only the final Cauchy support L is fixed. For a cylinder with smaller spatial radius $\ell = L - T$, the bound remains valid, but reachability of its top eigenfunction is not asserted. Exact complete dephasing at finite a , T , and R requires $E_K = \infty$.

Remark 3.7. The final statement is a no-go only for exact complete dephasing in the frozen local scalar model. It is not a universal energy cost for quantum measurement, and it does not price the external device used to realize J .

4 Optimal profiles and smooth source realization

4.1 Explicit trial profiles

The exact optimizer in theorem 3.5 is the top eigenfunction of $\mathcal{K}_{y/2}$. Two elementary profiles give the explicit lower bounds in equation (51) and make both asymptotic regimes concrete. For the vacuum-controlled regime, set $L = yR$, choose $q = 0$, and use

$$p_L(x) = \sqrt{\frac{3}{L^3}} x \mathbf{1}_{(0,L)}(x). \quad (60)$$

The vacuum part of the thermal form is

$$\begin{aligned} \langle p_L, h_0^{-1} p_L \rangle &= \frac{3}{\pi L^3} \int_0^L \int_0^L xy \log \frac{x+y}{|x-y|} dx dy \\ &= \frac{3L}{2\pi}. \end{aligned} \quad (61)$$

For the second equality, scale $x = Lu$, $y = Lv$, split the unit square across the diagonal, and use

$$\int_0^1 t \log \frac{1+t}{1-t} dt = 1. \quad (62)$$

Since $\coth(\beta h_0/2) \geq 1$, the full thermal dephasing exponent is at least the value in equation (61). On the other hand, $E_K = 1/2$ for the normalized datum. Thus $\Gamma/(E_K R) \geq 3L/(\pi R) = 3y/\pi$.

For the thermal regime, use instead the normalized first eigenfunction of the Dirichlet–Neumann Green operator,

$$p_L^{\text{th}}(x) = \sqrt{\frac{2}{L}} \sin \frac{\pi x}{2L} \mathbf{1}_{(0,L)}(x). \quad (63)$$

Since $\coth z \geq z^{-1}$ and $\langle p_L^{\text{th}}, h_0^{-2} p_L^{\text{th}} \rangle = 4L^2/\pi^2$, this datum gives

$$\frac{\Gamma}{E_K R} \geq \frac{8y^2}{\pi^3}. \quad (64)$$

The first profile and the vacuum Schur bound also show

$$\frac{3}{\pi} \leq \liminf_{y \downarrow 0} \frac{C_{\text{opt}}(y)}{y} \leq \limsup_{y \downarrow 0} \frac{C_{\text{opt}}(y)}{y} \leq \frac{4 \operatorname{arsinh}(1)}{\pi}. \quad (65)$$

The ratio of the analytic upper and lower linear coefficients is $4 \operatorname{arsinh}(1)/3 \simeq 1.17516$.

For a target exponent $\Gamma_\star$, scaling equation (60) by an amplitude A yields the sufficient energy bound

$$E_K \leq \frac{\pi \Gamma_\star}{3L}. \quad (66)$$

This is an upper bound on the energy needed by this simple family. The sharp half-line momentum infimum at inverse temperature β is instead $\Gamma_\star/C_\beta(L) = \Gamma_\star/[2L\Lambda(\pi L/\beta)]$. At the de Sitter temperature this is $\Gamma_\star/[RC_{\text{opt}}(L/R)]$.

4.2 Smooth compact sources

The hard endpoint in equation (60) is useful for exact constants but is not a smooth source. We now show that it is a legitimate limiting profile and that every strict window persists.

Lemma 4.1 (Smooth density closure). *Fix an optical source radius $\ell > 0$ and a normalized $p \in L^2(0, \ell)$. There are normalized $p_n \in C_c^\infty([0, \ell))$, regular at the origin, such that*

$$p_n \longrightarrow p \quad \text{in } L^2(\mathbb{R}_+). \quad (67)$$

Their thermal dephasing forms converge, and their cumulative energy measures converge uniformly.

Proof. Density of smooth functions supported strictly inside $(0, \ell)$ gives equation (67); such functions are automatically regular radial data at the origin. On data supported in $[0, \ell]$,

$$P_\ell h_0^{-1} \coth(\beta h_0/2) P_\ell \leq P_\ell h_0^{-1} P_\ell + \frac{2}{\beta} P_\ell h_0^{-2} P_\ell \quad (68)$$

as quadratic forms. The right side is bounded by lemma 3.1; hence the covariance is continuous under L^2 convergence. Finally,

$$\sup_{0 \leq x \leq \ell} \left| \int_0^x (p_n^2 - p^2) \, ds \right| \leq \|p_n - p\|_2 (\|p_n\|_2 + \|p\|_2), \quad (69)$$

which proves uniform convergence of cumulative energy. $\square$

It remains to produce a supported final datum by a smooth spacetime source. This also proves sharpness over sources when the source ball approaches the declared final-support ball; it does not prove exact controllability from a strictly smaller cylinder of optical radius $L - T$. Let $\phi_{\text{free},n}$ be the homogeneous solution with final data $(q, p) = (0, p_n)$. Choose a smooth time cutoff η_n that vanishes before the protocol, equals one near the final slice, and changes over a time shorter than the optical gap between $\text{supp } p_n$ and ℓ . If $\mathcal{P}$ is the conformal wave operator, set

$$J_n = \mathcal{P}(\eta_n \phi_{\text{free},n}). \quad (70)$$

The retarded solution sourced by J_n is $\eta_n \phi_{\text{free},n}$. It therefore has exactly the desired final data. Finite propagation keeps J_n inside the declared optical worldtube. This construction proves smooth source existence, but it does not bound the stress of an autonomous mechanism that generates the prescribed J_n .

5 Relation to prior bounds and detector work

5.1 Bekenstein entropy is a different quadratic form

For a Klein–Gordon wave packet localized in a spatial region of half-width a , Hollands, Longo, and Morsella prove the Bekenstein-type inequality

$$S(\Phi \mid B) \leq 2\pi a E, \quad (71)$$

where $S(\Phi \mid B)$ is the entropy quadratic form of the local standard subspace [9]. At second quantization this is the relative entropy of the corresponding coherent excitation. This powerful result is directly adjacent to theorem 3.3, but it controls a different functional.

We make the distinction explicit in the simplest common setting. Consider the massless conformal vacuum in four-dimensional Minkowski space, restrict to radial $q = 0$ data in a ball of radius a , and normalize $\|p\|_2 = 1$. The conformal-ball modular Hamiltonian and the half-line dephasing kernel give

$$S_B[p] = \frac{\pi}{2a} \int_0^a (a^2 - x^2) p(x)^2 dx, \quad (72)$$

$$\Gamma_0[p] = \langle p, h_0^{-1} p \rangle = \frac{1}{\pi} \int_0^a \int_0^a p(x) p(y) \log \frac{x+y}{|x-y|} dx dy. \quad (73)$$

Proposition 5.1 (Quadratic-form separation). *At fixed a , there is no finite constant c such that $\Gamma_0[p] \leq c S_B[p]$ for all smooth normalized radial momenta, and no finite constant c' such that $S_B[p] \leq c' \Gamma_0[p]$ for all such data.*

Proof. Let $f \in C_c^\infty(0, 1)$ be positive and normalized in $L^2(0, 1)$. For a packet in a boundary layer, set

$$p_\delta^{\text{wall}}(x) = \delta^{-1/2} f\left(\frac{a-x}{\delta}\right). \quad (74)$$

Substitution in equation (72) gives

$$S_B[p_\delta^{\text{wall}}] = \pi \delta \int_0^1 u f(u)^2 du + O(\delta^2). \quad (75)$$

For the logarithmic kernel, scaling both variables gives

$$\Gamma_0[p_\delta^{\text{wall}}] = \frac{\delta}{\pi} \left(\int_0^1 f(u) du \right)^2 \log \frac{1}{\delta} + O(\delta). \quad (76)$$

Thus $\Gamma_0/S_B \rightarrow \infty$.

For a packet near the center, set

$$p_\delta^{\text{ctr}}(x) = \delta^{-1/2} f(x/\delta). \quad (77)$$

Then

$$S_B[p_\delta^{\text{ctr}}] = \frac{\pi a}{2} + O(\delta^2), \quad \Gamma_0[p_\delta^{\text{ctr}}] = O(\delta), \quad (78)$$

because the logarithmic kernel becomes scale invariant after $x = \delta u$, $y = \delta v$, while the two integration measures leave one factor of δ . Hence $S_B/\Gamma_0 \rightarrow \infty$. $\square$

The separation has a simple physical meaning. The Bekenstein entropy is a local modular-information functional whose conformal weight vanishes at the ball boundary. The zero-temperature dephasing exponent is a global one-particle norm, equivalently an entangling-particle count for the exact detector model. At finite temperature, the factor $\coth(\beta h/2)$ enhances infrared fluctuations further. Thermal dephasing can increase even while the distinguishability of two displaced thermal states decreases. The two notions of information should not be identified.

5.2 Relation to DSW and exact detector work

DSW decoherence describes the loss of coherence of superposed charges or masses near Killing horizons through soft field modes [4, 6]. Gralla and Wei gave a general horizon formulation and precise rates in electromagnetic and Klein–Gordon analogues [5]; Li evaluated the local scalar, electromagnetic, and gravitational cases in de Sitter [7]. These works establish the physical importance of the low-frequency KMS sector that appears in equation (13).

Our setup differs in its question. We do not derive a new horizon decoherence rate. We start from the exact pointer channel and maximize its dephasing functional subject to finite final support and post-switch field energy. The result is consequently a constraint on a local implementation, not a claim that horizon modes require a large Killing energy. Indeed, sufficiently soft horizon radiation can carry negligible Killing energy when its support and duration are allowed to grow. In equation (59), that escape route is visible in the growth of C_{opt} with $L = \ell + T$.

Barcellos and Landulfo already separate field background energy, individual switching work, and communication work for the exact gapless-detector channel [2]. We use their same exact channel class. The new step is to eliminate the source amplitude, identify the exact compact KMS kernel, and solve the full final-support optimization. We therefore make no novelty claim for Magnus termination, the KMS characteristic function, the coherent stress identity, or switching-energy accounting.

Likewise, spectral extremization of Riesz-potential operators is not new [8]. In the zero-temperature limit our reflected logarithmic kernel belongs to that broad potential-theoretic family. The claim here is not a first eigenvalue theorem for a positive compact kernel; it is the identification of the exact thermal operator governing this local pointer channel, proof that it controls the complete canonical and angular phase space, and extraction of the sharp physical support laws.

6 Discussion and claim boundary

The main result is an exact final-support optimization, not a new detector channel. At arbitrary inverse temperature, the half-line momentum problem is governed by the single compact operator $\mathcal{K}_{\pi L/\beta}$. In the conformal de Sitter model, angular resolvent order and a separate field-coordinate estimate show that this momentum problem is the full canonical and angular optimum. The temperature-curvature relation is used in that second step, not in the general half-line theorem.

Compact final support is the decisive hypothesis. Without it, arbitrarily soft thermal modes prevent an energy norm from controlling the dephasing exponent. With it, the compressed inverse Hamiltonian is bounded and the exact coefficient has both a vacuum-controlled small-support regime and a thermal large-support regime. The uniform remainder estimates in theorem 3.3 make those statements quantitative.

6.1 What is sharp

The results establish:

1. the exact arbitrary- β half-line momentum coefficient $C_\beta(L) = 2L\Lambda(\pi L/\beta)$, its unique optimizer, global bracket, and support asymptotics;
2. the conformal de Sitter reduction of every angular and canonical sector to the unique s-wave momentum optimizer;
3. the exact pointer channel and the post-switch scalar-field energy and renormalized-stress excess generated by the same prescribed source;
4. a causal source-radius and duration envelope for the final support; and
5. smooth compact sources approaching every final-support Rayleigh quotient when the source worldtube approaches the final-support ball.

Sharpness refers to the optimization over final Cauchy data supported in $[0, L]$. For a source cylinder with spatial radius $\ell < L$ and duration $T = L - \ell$, finite propagation proves the same upper bound but

not near-controllability of its top eigenfunction. The title and main theorem do not assert that stronger statement.

The field-energy quantity is equally specific. It is the post-switch scalar Killing energy of the coherent displacement. It does not include the work, clock, battery, trap, or material stress needed to realize the externally prescribed source. The smeared qubit is a standard gapless-detector idealization rather than an autonomous relativistic probe field. Accordingly, the theorem is not a universal measurement-energy principle.

6.2 Novelty boundary

Exact gapless-detector channels, their thermal damping factors, and their field-energy accounting are prior art. Positive compact kernels, simple top eigenvalues, and Riesz-potential variational principles are also established mathematics. The candidate contribution is their specific conjunction here: the reflected thermal kernel selected by final support, the exact general-temperature coefficient, the full de Sitter sector reduction, and the uniform support asymptotics. Independent review in both relativistic detector theory and integral-operator or fractional-Sobolev theory remains the binding publication gate.

The localized Klein–Gordon entropy comparison in section 5 is supporting evidence that the dephasing quadratic form is not a repackaging of a recent Bekenstein bound. It is not a second headline. Likewise, appendix A is an appendix application to final-slice constraint data. It does not derive the pointer channel on a perturbed geometry, include actuator stress, or solve a coupled Einstein–matter evolution.

6.3 Extensions

The most direct mathematical extension is the true fixed-source-cylinder optimization. It requires either a near-controllability theorem for compact sources in a prescribed smaller worldtube or a new source-to-final-data operator whose singular value replaces Λ . A broader geometric extension would identify conditions under which localized norms of $h^{-1} \coth(\beta h/2)$ admit comparable sharp constants on other static spacetimes.

The cleanest physical extension is to replace the shared smeared qubit by a localized relativistic probe field, following the detector-from-field framework of [11]. Such a model would have to retain the localization theorem while accounting for additional probe modes and their stress tensor. Neither extension is required for the narrow theorem proved here.

6.4 Reproducibility

From the repository root, the theorem record, numerical illustration, and focused checks are reproduced by

```
PYTHONPATH=. python experiments/local_scalar_observer_cost_audit.py
PYTHONPATH=. python experiments/local_scalar_observer_spectrum.py
PYTHONPATH=. python -m pytest -q \
    tests/test_local_scalar_observer_cost.py \
    tests/test_local_scalar_observer_manuscript.py \
    tests/test_local_scalar_observer_spectrum.py
```

The frozen records check formulas, source hashes, numerical convergence, and artifact provenance. They do not replace the analytic proof or certify literature novelty.

A Final-slice Einstein–scalar application

This appendix is not part of the paper’s headline theorem. The all-angular de Sitter result does not require spherical symmetry. We now restrict to the s-wave and to a final slice on which the radial field coordinate vanishes. This subclass is useful because its momentum density vanishes and its energy measure has a direct constraint interpretation.

Let $u(t, x)$ be the normalized rescaled s-wave. The physical conformal scalar is

$$\phi(t, r) = \frac{u(t, x(r))}{\sqrt{4\pi} r}. \quad (79)$$

On a final slice with $u = 0$ and $p = \partial_t u$, the physical normal derivative in the de Sitter geometry is

$$v(r) := n^a \nabla_a \phi = \frac{p(x(r))}{\sqrt{4\pi} r \sqrt{N(r)}}. \quad (80)$$

For the conformally coupled stress tensor, every improvement term containing ϕ or a spatial derivative of ϕ vanishes on this slice. Directly,

$$\rho = T_{ab} n^a n^b = \frac{1}{2} v^2, \quad j_i = -T_{ab} n^a h^b_i = 0. \quad (81)$$

Since $dr = N dx$, equations (80) and (81) imply the exact measure identity

$$4\pi r^2 \rho(r) dr = \frac{1}{2} p(x)^2 dx. \quad (82)$$

Use the same numerical function v as the scalar normal-momentum datum on a new spherical initial slice, and consider gravitational data with vanishing extrinsic curvature,

$$dl^2 = \frac{dr^2}{f(r)} + r^2 d\Omega_2^2, \quad K_{ij} = 0, \quad f(r) = N(r) - \frac{2Gm(r)}{r}. \quad (83)$$

The momentum constraint is satisfied by equation (81). The Hamiltonian constraint with cosmological radius R is solved by

$$m'(r) = 4\pi r^2 \rho(r), \quad m(0) = 0. \quad (84)$$

Absorbing the Bunch–Davies vacuum expectation into the background cosmological constant, the coherent excess stress in equation (21) supplies precisely the classical source above. Although $K_{ij} = 0$, the nonzero scalar normal momentum means that the full Einstein–scalar data are not invariant under time reflection. Equation (82) then gives

$$m(b) = E_K \quad (85)$$

for any areal radius b outside the field support.

Define the dimensionless radial constraint ratio

$$Q(r) = \frac{2Gm(r)}{rN(r)}. \quad (86)$$

For a comparison ball $0 < r \leq b < R$, set

$$Q_b := \sup_{0 < r \leq b} Q(r). \quad (87)$$

If $Q_b \leq \delta < 1$, then throughout that ball

$$f(r) \geq N(r)(1 - \delta) > 0, \quad 0 \leq \frac{g_{rr}}{g_{rr}^{\text{dS}}} - 1 \leq \frac{\delta}{1 - \delta}. \quad (88)$$

Corollary A.1 (Necessary final-slice constraint cost). *Let the hypotheses of corollary 3.6 hold and suppose the source is engineered to finish with spherical $q = 0$ data. Set*

$$b = R \tanh(L/R), \quad L = R \operatorname{arctanh}(a/R) + T. \quad (89)$$

Then

$$Q_b \geq Q(b) \geq \frac{2G}{bN(b)RC_{\text{opt}}(L/R)} \log \frac{1}{2\epsilon_{\text{obs}}}. \quad (90)$$

Equivalently, a local budget $Q_b \leq \delta < 1$ implies

$$\epsilon_{\text{obs}} \geq \frac{1}{2} \exp \left[-\frac{\delta bN(b)R}{2G} C_{\text{opt}}(L/R) \right]. \quad (91)$$

Proof. Outside the support, equation (85) gives $Q(b) = 2GE_K/[bN(b)]$. Apply the sharp energy lower bound in corollary 3.6; solving the same inequality in the other direction gives equation (91). $\square$

The compact mass shifts the cosmological horizon relative to the pure de Sitter value R . Accordingly, Q_b is a local comparison norm on the ball containing the final datum; no global smallness claim relative to the unperturbed lapse is intended.

A.1 An explicit local weak-constraint window

For the source optical radius $\ell = R \operatorname{arctanh}(a/R)$, take the ideal profile

$$q = 0, \quad p(x) = A \sqrt{\frac{3}{\ell^3}} x \mathbf{1}_{(0,\ell)}(x). \quad (92)$$

Its cumulative mass is

$$m(x) = E_K \left(\frac{x}{\ell} \right)^3. \quad (93)$$

Inside the support, the function

$$\frac{\operatorname{arctanh}(r/R)^3}{r[1 - r^2/R^2]} \quad (94)$$

is strictly increasing on $(0, a]$. One way to see this is to factor it as $(x/r)(x^2/N)$; both factors increase with r . Hence the wall value controls the complete profile. Combining equation (66) with the wall constraint gives the sufficient estimate

$$Q_a \leq \frac{2\pi G \Gamma_\star}{3\ell a[1 - a^2/R^2]}. \quad (95)$$

For illustration, set

$$\epsilon_{\text{obs}} = 10^{-6}, \quad a/R = 0.2, \quad T/R = 0.1, \quad G/R^2 = 10^{-6}, \quad \delta = 0.25. \quad (96)$$

The universal theorem requires

$$C_{\text{opt}}(L/R) \leq 0.340766570374, \quad E_K R \geq 38.5083647231, \quad Q(b) \geq 2.86896268303 \times 10^{-4}, \quad (97)$$

whereas the constructive family has

$$E_K R \leq 67.7824380936, \quad Q_a \leq 7.06067063475 \times 10^{-4}. \quad (98)$$

Thus the fixed numerical point has a nonempty local weak-constraint window. By lemma 4.1, smooth compact profiles preserve this strict margin.

The construction closes only the final Hamiltonian and momentum constraints. It does not specify the lapse, evolve the Einstein–scalar system, include the source actuator during switching, or recompute the quantum channel in the perturbed geometry.

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